

BS-Regular (CHEM-422) (Past Papers)
Nuclear Chemistry (CHEM-449)

Short Questions:

- 1-Explain why the mass of an atom is less than the sum of its component particles?
- 2-Differentiate between Natural and Artificial radioactivity.
- 3-Explain the uses of radioactive traces in analytical chemistry.
- 4-Write about harmful effects of radioactivity.
- 5-Explain the mechanism of Stellar Fusion reaction of Hydrogen.
- 6-What is the function of moderator in Nuclear Chemistry?
- 7-What are the units of radiation exposure?
- 8-Describe three research applications of traces in Chemistry.

Long Questions:

- 1-What is Radioactivity? Describe the rates of Disintegration of radioactive elements.
- 2-Distinguish between Fission and Fusion with the help of examples.
- 3-Write a note on the following:
 - (a)Nuclear reactor
 - (b)Stellar energy
 - (c)radioactive equilibrium
- 4-Explain decay constant with its units.
- 5-Derive an equation to calculate Half-life of a radioactive substance.
- 6-Write a detailed note on the Atomic Bomb.
- 7-Explain the radiation Hazards in detail.

M.Sc.-II (Past Papers)

Answer the following questions:

- 1-What are accelerators? Give basic principle of accelerators. Give an account of linear accelerator. (2015)
- 2-What is a nuclear reactor? Give its basic components. Explain any one of the power reactor in detail. (2015)

- 3-Write a note on Artificial radioactivity. (2015)
- 4-What are tracers?Highlight Applications of tracers in chemistry. (2016)
- 5-What is radioactive disintegration?Explain briefly disintegration as a first order process.(2016)
- 6-How can you find the half-life period of a radioactive decay? (2016)
- 7-Discuss the details of nuclear fission and fusion processes. (2016,2018)
- 8-Write a note on radioactive equilibrium. (2016,2018,2022)
- 9-What is radioactivity?Discuss the kinetics of disintegration of a radioactive substance.(2017,2020)
- 10-Prove that the Half-life period of a radioactive substance is independent of its initial atoms. (2017,2020)
- 11-What are tracers?Discuss their principles and applications. (2017,2020,2022)
- 12-What is nuclear decay?Discuss the kinetics of nuclear decay. (2018,2022)
- 13-Differentiate between Nuclear and Chemical reactions.Discuss the use of traces in Chemistry. (2018)
- 14-Distinguish between Nuclear fission and nuclear fusion with the help of examples.Which the process is more energetic and why? (2019)
- 15-What is artificial radioactivity?Describe the rate of disintegration of radio-active elements.Give examples. (2019)
- 16-What is radioactive decay?Give its types of radioactive decay with examples.Discuss the kinetics of radioactive decay in detail. (2021)
- 17-What are nuclear processes?Discuss nuclear fission and nuclear fusion in detail.(2021,2022)